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DeepFinLLM: An Intelligent Financial Advisor Unleashing Strategic Insights with Large Language Models

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Abstract

Generative AI breakthroughs have reshaped financial analytics, introducing seamless, language-based interactions that expand access to valuable insights. Traditionally, financial analysis relied on rigid, rule-driven tools requiring specialized skills, often yielding 75% accuracy and faltering with complex queries. DeepFinLLM overcomes these barriers by blending advanced language understanding with real-time financial data, offering precise responses with a 94% accuracy rate and 0.8-second delivery time. Merging artificial intelligence with dynamic market data, it provides adaptability and scalability across diverse financial contexts. Users, from expert investors to novices, praise its intuitive interface, with 90% valuing its fast, reliable insights. Handling structured financial information and open-ended inquiries, DeepFinLLM facilitates deep analysis of investment options, market trends, and risks, enhanced by real-time updates for relevance in volatile markets. By simplifying complex tasks and encouraging data-informed decisions, DeepFinLLM transforms financial analytics to serve a broader audience effectively.

Keywords: Large Language Models (LLMs), Financial Analytics, Strategic Insights, Natural Language Processing (NLP)

1 Introduction

The rapid advancement of artificial intelligence (AI) and NLP has fundamentally reshaped how machines interpret and respond to complex human queries. In recent years, LLMs have showcased exceptional abilities in processing intricate datasets and generating contextually relevant insights across various domains, including financial analytics, market forecasting, and investment decision-making. However, delivering precise, real-time financial insights requires effective integration of dynamic market data and semantic understanding, particularly when handling vast, volatile financial corpora. Traditional financial analysis tools, reliant on rule-based algorithms and sparse statistical methods, often fail to capture the nuanced dynamics of market trends, achieving accuracy rates of only 70% and struggling with multi-dimensional queries (Adams & Lee, 2022). Similarly, while modern dense retrieval techniques using vector-based representations enhance semantic matching, they may overlook critical real-time updates, reducing their effectiveness in fast-paced financial environments.

To address these shortcomings, hybrid systems that combine the strengths of traditional rule-based approaches and advanced AI-driven methods have emerged as a promising solution. By integrating lexical precision with semantic depth and real-time data, such systems achieve superior accuracy and responsiveness, essential for delivering actionable financial insights. Additionally, the incorporation of adaptive data processing techniques has improved contextual relevance by dynamically aligning live market feeds with user queries, enhancing both precision and analytical quality (Carter & Nguyen, 2023). Despite these advancements, challenges persist in efficiently merging real-time data with LLM outputs and evaluating the reliability of financial insights, particularly in volatile market conditions.

In our proposed system, we introduce DeepFinLLM, an intelligent financial advisor that combines sparse rule-based analysis with dense LLM-driven retrieval, utilizing a fusion strategy to balance accuracy and timeliness. The system leverages the DeepSeek Chat V3 model for semantic processing and the Financial Modeling Prep (FMP) API for real-time data integration, ensuring that retrieved financial context is both relevant and current (Patel & Lee, 2022). Furthermore, the generated insights are assessed using a comprehensive evaluation framework, incorporating metrics such as accuracy (94.2%) for correctness, inference latency (0.8 seconds) for efficiency, and factual consistency (96.5%) for reliability. This multi-faceted evaluation approach provides a robust assessment of the quality and applicability of the financial advice generated.

One of the primary challenges in financial analytics lies in maintaining contextual accuracy and responsiveness across diverse market scenarios. Traditional tools often struggle with real-time adaptability, resulting in delayed or imprecise insights. DeepFinLLM tackles this by employing a hybrid retrieval strategy that captures both structured financial data and semantic market trends, ensuring that insights remain contextually faithful (Chang & Liu, 2024). However, existing financial advisory systems still face several limitations:

- **Limited real-time awareness:** Many systems rely on static data inputs, leading to outdated insights or incomplete market representations, which diminish decision-making quality (Hughes & Wang, 2022).

- **Inconsistent integration strategies:** While hybrid approaches improve accuracy, their effectiveness hinges on data fusion techniques that may not generalize across varied financial domains.
- **Evaluation complexity:** Assessing financial insights involves multiple dimensions accuracy, timeliness, and compliance yet existing metrics often fail to holistically capture these aspects.
- **Computational overhead:** Combining LLMs with real-time data streams introduces significant computational demands, potentially impacting scalability in high-volume applications (Nguyen & Taylor, 2025).

To overcome these limitations, DeepFinLLM introduces several key innovations. By merging sparse and dense retrieval with real-time data, the system ensures precise and contextually rich financial insights. Its adaptive data integration dynamically adjusts to market volatility, preventing loss of relevance due to static processing (Turner & Singh, 2022). Additionally, the multi-metric evaluation framework offers a comprehensive assessment of insight quality, making DeepFinLLM suitable for applications such as investment strategy formulation, risk assessment, and market trend analysis.

This paper is structured into seven pivotal sections. Section 2 provides a comprehensive background, tracing the financial sector’s evolution and technological challenges, including the limitations of traditional tools (Adams & Lee, 2022). Section 3 reviews related work, synthesizing financial analytics’ progression and AI-driven innovations, such as NLP-enhanced systems (Chang & Liu, 2024). Section 4 details the proposed system architecture, integrating LLMs with real-time data streams via APIs (Patel & Lee, 2022). Section 5 evaluates its performance against conventional and advanced models, using benchmarks and diverse datasets to validate metrics like accuracy and latency. Section 6 examines threats to validity and limitations, such as scalability concerns (Nguyen & Taylor, 2025), affecting broader applicability. Section 7 concludes with key insights and future research directions.

2 Background

The financial sector has undergone significant transformation due to technological advancements and the growing complexity of data, reshaping how investment decisions are made. Investors now face a diverse array of information stock prices, economic reports, social media trends, and regulatory updates requiring advanced tools to process this overload effectively (Baker & Singh, 2022). Financial advisory systems have become essential, offering structured approaches to portfolio optimization, risk assessment, and market prediction (Davis & Thompson, 2022). Artificial intelligence (AI) has driven this evolution, turning static tools into dynamic, intelligent platforms (Clark & Evans, 2022). *DeepFinLLM* leverages these developments, combining LLMs with real-time data to provide adaptive, user-oriented financial guidance. This section explores the historical context, the role of LLMs, and the importance of real-time data, setting the stage for our proposed system.

2.1 Historical Tools in Financial Analytics

Financial advisory systems originated with basic tools designed to aggregate and analyze market data. Early platforms provided foundational resources for professionals, relying on structured inputs like historical prices (Patel & Lee, 2022). The introduction of robo-advisors marked a shift, automating portfolio management with algorithms focused on risk-return balance (Brooks & Yang, 2022). These tools improved access to financial advice but faced difficulties adapting to sudden market changes (Deng & Zhou, 2023). Over time, systems evolved to incorporate predictive capabilities and interactive features, reflecting a move toward more sophisticated analytics (Kim & Taylor, 2023; Simmons & Zhang, 2023). Query processing interfaces emerged, enhancing efficiency, though scalability remained limited by static data reliance (Becker & Kim, 2023; James & Zhou, 2024; Reed & Patel, 2023). This progression laid the groundwork for integrating advanced technologies into financial decision-making.

2.2 Emergence of AI and Language Models

AI has redefined financial analytics by introducing systems capable of learning and reasoning. Initial advancements improved prediction accuracy using machine learning, particularly with structured data (Ahmed & Patel, 2023). The rise of LLMs brought new capabilities, leveraging NLP to interpret complex, unstructured sources like reports and news (Chang & Liu, 2024). Sentiment analysis and query resolution became possible, expanding the scope of financial applications (Jackson & Yang, 2023; Thomas & Patel, 2024). Dynamic uses, such as decision support, emerged, though challenges in processing speed and clarity persisted (Hill & Nguyen, 2024; Nelson & Wang, 2024; Phillips & Liu, 2024). These developments established LLMs as a cornerstone for modern financial tools, influencing *DeepFinLLM*'s design with DeepSeek Chat 3 to address these limitations.

2.3 Real-Time Data and Modern Finance

The need for immediate insights has made real-time data a critical component of financial systems, especially in fast-paced markets (Hughes & Wang, 2022). Older tools, dependent on delayed updates, often missed rapid shifts, reducing their reliability (Adams & Lee, 2022). Integrating live data streams stock quotes, forex rates, and economic indicators became essential for timely decisions (Turner & Singh, 2022). Precision improvements followed, driven by AI systems processing real-time inputs (Chen & Wu, 2025). Innovations in market analysis and automation further highlighted this trend, though user accessibility lagged (Edwards & Kim, 2024; Lopez & Zhang, 2024; Moore & Singh, 2023; Taylor & Kim, 2023). *DeepFinLLM* builds on this by pairing the Financial Modeling Prep (FMP) API with LLM analysis, aiming for both timeliness and ease of use (Scott & Liu, 2025). This convergence of historical tools, AI advancements, and real-time data underpins our approach to intelligent financial advisory systems.

3 Related Work

The development of financial advisory systems has been shaped by extensive research addressing prediction accuracy, real-time processing, language comprehension, and user accessibility. Unlike earlier tools, modern systems aim to handle complex, dynamic markets, a challenge *DeepFinLLM* tackles through integrated LLMs and real-time data streams. This section examines prior work across key areas: traditional forecasting, AI-driven analytics, LLM applications, real-time integration, and user-focused design, comparing their contributions and limitations to highlight how *DeepFinLLM* advances the field.

3.1 Traditional Forecasting and Optimization Systems

Early financial advisory systems focused on structured data to predict trends and optimize portfolios. Predictive models achieved moderate success, with accuracy rates around 70–80%, using historical prices and economic indicators (Price & Kim, 2025). Portfolio optimization relied on statistical techniques to balance risk and return, though adaptability to new data types was limited (Liu & Patel, 2023). Accessibility issues arose due to expert-dependent configurations, reducing their reach among non-specialists (Harris & Kim, 2022). Delayed responses during market volatility further exposed rigidity, with accuracy dropping in complex scenarios (Adams & Lee, 2022). While these systems established quantitative baselines, their inability to process unstructured inputs or respond swiftly left significant gaps, which *DeepFinLLM* addresses with dynamic, language-driven capabilities.

3.2 AI-Driven Financial Analytics

The shift to artificial intelligence introduced adaptive analytics through machine learning (ML). Market trend forecasting improved incrementally over statistical methods by leveraging historical datasets, though real-time integration lagged due to processing delays (Quinn & Zhang, 2022). Risk evaluation gained precision with supervised learning, but manual feature engineering restricted flexibility with diverse data (Russell & Yang, 2023). Real-time decision-making systems faced similar complexity hurdles, requiring technical expertise that limited practical use (Ross & Lee, 2022). Scalable ML analytics handled larger datasets, enhancing predictive power, yet user-friendliness remained a challenge (Kim & Taylor, 2023). These efforts showcased AI’s potential, but their focus on structured data and back-end efficiency contrasts with *DeepFinLLM*’s emphasis on unstructured processing and accessibility.

3.3 Applications of LLMs

LLMs brought NLP to financial analytics, enabling new interaction modes. Sentiment analysis accuracy improved significantly by extracting insights from unstructured texts like earnings reports, surpassing traditional NLP tools (Chang & Liu, 2024). Query resolution systems generated coherent responses from diverse inputs, though latency hindered time-sensitive use (Shaw & Patel, 2022). Financial analytics models extracted actionable insights from complex documents, but static analysis dominated

over dynamic applications (Thomas & Patel, 2024). Advisory systems saw engagement boosts from conversational LLMs, yet static training data limited market responsiveness (King & Wu, 2024). Versatile query handling emerged, though high-stakes scenarios revealed latency and interpretability issues (Hill & Nguyen, 2024); (Phillips & Liu, 2024). Decision support systems enhanced forecasting flexibility, underscoring the need for real-time data (Nelson & Wang, 2024). Compared to these niche-focused LLMs, *DeepFinLLM* integrates broader, real-time capabilities with DeepSeek Chat V3 for faster, more reliable outputs.

3.4 Real-Time Data Processing Systems

Real-time data integration became critical as markets demanded immediacy. Risk assessment models reduced latency with live feeds, though language interfaces were absent, limiting usability (Bell & Torres, 2024). ML-driven advisory systems scaled with dynamic streams, but structured input reliance constrained scope (Peters & Yang, 2023). High-frequency trading gained a competitive edge with live analytics, prioritizing efficiency over accessibility (Smith & Taylor, 2024; Moore & Singh, 2023). Automation improved through real-time AI pipelines, delivering precision in volatile conditions, though resource demands were high (Lopez & Zhang, 2024). Market analysis tools achieved rapid updates with generative AI, targeting expert users rather than novices (Edwards & Kim, 2024). Precision gains were notable in real-time AI systems, yet intuitive design lagged (Chen & Wu, 2025; Cruz & Diaz, 2025). *DeepFinLLM* combines the Financial Modeling Prep (FMP) API with LLM analysis, overcoming these trade-offs to offer both speed and user accessibility.

3.5 User-Centric Design in Advisory Tools

Enhancing usability has been a growing focus in financial systems. Scalable AI architectures personalized investment advice, boosting adoption rates over traditional tools, though dynamic data was lacking (Alvarez & Gomez, 2024). Hybrid systems blended ML and basic NLP for planning accuracy, constrained by static inputs (Barnes & Chen, 2025). Trust among non-experts improved with accessible designs, but real-time integration was absent (Rogers & Wu, 2025). Intuitive interfaces increased engagement in AI advisors, often without advanced LLMs (Garcia & Patel, 2025). User-centric generative AI emerged, though scalability and real-time features were underdeveloped (Spencer & Wu, 2025). Large-scale risk management systems addressed scalability, but domain-specific focus limited versatility (Nguyen & Taylor, 2025). Query processing efficiency rose with language interfaces, and scalable advisors evolved, yet static data persisted as a bottleneck (Becker & Kim, 2023; James & Zhou, 2024). *DeepFinLLM* integrates these user-focused advances with real-time, LLM-driven insights, broadening applicability.

3.6 Positioning DeepFinLLM

DeepFinLLM distinguishes itself by synthesizing and surpassing prior work. Traditional systems struggled with rigid rules and static data, while early AI lacked language proficiency, gaps filled by LLM-driven unstructured query processing. Latency and

real-time challenges in LLM and risk models are mitigated with the FMP API for timely updates. Unlike efficiency-focused real-time tools or static advisory systems, it balances responsiveness and accessibility. Interpretability and latency hurdles are addressed with DeepSeek Chat V3, enhancing strategic insights across users. Building on user-centric designs and real-time capabilities, it overcomes scalability, accessibility, and adaptability limitations, advancing intelligent financial advisory systems.

4 Methodology

This section outlines the methodology of DeepFinLLM, a sophisticated web-based system engineered to transform natural language financial queries into precise, data-driven insights and actionable recommendations. The methodology encompasses a multi-layered system architecture, advanced NLP techniques, real-time data integration, a robust financial insight generation mechanism, user customization strategies, and a comprehensive evaluation process. Each component is meticulously designed to address the complexities of interpreting user intents, interfacing with live financial data, and delivering decisive insights tailored to diverse financial contexts.

4.1 System Architecture

The system architecture of DeepFinLLM is a multi-layered design, engineered to process financial queries efficiently and deliver actionable insights with scalability and robustness. It consists of five interdependent layers: Presentation, State Management, Data Processing, API Integration, and Persistence. Each layer serves a distinct purpose in the query-to-insight pipeline, ensuring modularity and flexibility for independent optimization while maintaining overall system coherence. The architecture is designed to handle real-time financial demands, with performance metrics modeled mathematically to guarantee responsiveness, context retention, and data integrity.

4.1.1 Presentation Layer

The Presentation Layer serves as the user interface, accepting natural language queries and presenting insights in an accessible format. It is optimized for responsiveness and usability across devices, catering to diverse users, from financial novices to experts. Its performance is quantified through efficiency metrics that balance query input processing and insight rendering. This layer captures user inputs, such as “What’s the risk of investing in tech stocks now?”, and forwards them to the State Management Layer. It also displays responses textual insights or visualizations like charts generated by downstream layers. Responsiveness is critical for real-time financial applications, particularly in volatile markets. The efficiency of the Presentation Layer is defined as:

$$E_{\text{Pres}} = \frac{N_{\text{queries}}}{T_{\text{render}} + T_{\text{input}}} \quad (1)$$

where N_{queries} is the number of processed queries, T_{render} the average time to render insights, and T_{input} is the input processing time. Achieving high throughput (e.g., rendering insights in under 0.5 seconds) ensures seamless interaction, critical in volatile

markets. The layer’s modular structure allows updates like adding new chart types without affecting downstream components, enhancing adaptability.

4.1.2 State Management Layer

The State Management Layer maintains conversational continuity by tracking query history and context, enabling iterative dialogues (e.g., “How does that compare to last year?”). It uses a finite state machine to manage states, ensuring context preservation across sessions. Its performance is measured through memory retention and state transition accuracy. The finite state machine defines states S_t that encapsulate current context, including prior queries and user preferences. State transitions occur as:

$$S_{t+1} = f(S_t, Q_t, C_t) \quad (2)$$

where S_t is the current state, Q_t the new query at time t , C_t includes contextual data (e.g., prior responses), and f is the transition function. Performance hinges on memory retention and state transition accuracy, ensuring context isn’t lost during multi-turn interactions. This layer supports real-time adaptability by caching recent states, reducing latency in follow-up queries.

4.1.3 Data Processing Layer

The Data Processing Layer is the analytical core, integrating NLP outputs with real-time data to generate insights. It balances high throughput and low latency to deliver near real-time responses. Performance is quantified through throughput and processing time metrics. This layer processes structured query representations and live data using LLMs. The throughput is:

$$T_{DP} = \frac{D_{\text{processed}}}{T_{\text{compute}} + T_{\text{integrate}}} \quad (3)$$

where $D_{\text{processed}}$ represents data volume per query, T_{compute} computation time, and $T_{\text{integrate}}$ data integration time. Parallel processing reduces the average response time to 0.8s, significantly faster than traditional systems’ minute-scale delays, enhancing real-time applicability.

4.1.4 API Integration Layer

The API Integration Layer fetches live financial data (e.g., stock prices, economic indicators) asynchronously from external sources. It ensures data timeliness and minimizes latency, with efficiency modeled through fetch rates and latency metrics. Asynchronous API calls enable concurrent data retrieval. The fetch rate is:

$$R_{API} = \frac{\Delta D_{\text{live}}}{\Delta t_{\text{fetch}}} \quad (4)$$

where ΔD_{live} is the fetched data volume and Δt_{fetch} the fetch interval. Asynchronous operations allow concurrent retrieval of multiple data points, reducing wait times to

under 0.3 seconds per call. A caching mechanism stores frequently accessed data, further optimizing performance while maintaining freshness through user-defined refresh rates.

4.1.5 Persistence Layer

The Persistence Layer stores query logs, user preferences, and historical insights for personalization and analysis. It ensures data integrity and fast access, with efficiency metrics for storage and retrieval. Storage efficiency is:

$$E_{\text{Pers}} = \frac{S_{\text{stored}}}{T_{\text{access}} + T_{\text{write}}} \quad (5)$$

where S_{stored} is the stored data volume, T_{access} retrieval time, and T_{write} write time. A flexible schema accommodates diverse data types text, numerical metrics, user settings enabling personalization and long-term analysis. Fast retrieval (e.g., under 0.1 seconds) supports real-time customization and auditability. This architecture ensures scalability and efficiency of DeepFinLLM, integrating all layers into a cohesive system for real-time financial analytics. The insight generation process is modeled as:

$$\text{Insight} = E(U(G(R(N(Q), D_{RT})), P_U), M, B) \quad (6)$$

where Q is the user query, $N(Q)$ NLP processing (converts Q to structured form S), $R(S, D_{RT})$ adds real-time data D_{RT} (e.g., FMP API data), $G(R(S, D_{RT}))$ generates insight I using DeepSeek Chat V3, $U(I, P_U)$ customizes I with user preferences P_U , and $E(U(I), M, B)$ evaluates the result with metrics M (e.g., accuracy) and baselines B .

4.2 NLP Processing for Query Analysis

The NLP pipeline of DeepFinLLM is engineered to dissect natural language queries, converting them into structured representations for financial analysis. It employs a sequence of techniques: tokenization, part-of-speech (POS) tagging, named entity recognition (NER), and intent classification. Tokenization breaks queries into discrete units (e.g., “What’s the trend for Tesla stock in 2025?” splits into “trend,” “Tesla,” “2025”). POS tagging assigns roles (e.g., “trend” as noun), NER identifies entities (e.g., “Tesla” as company), and intent classification determines goals using:

$$\text{intent} = \arg \max_i P(\text{class}_i \mid \text{query}, \theta) \quad (7)$$

where $P(\text{class}_i \mid \text{query}, \theta)$ is the probability of intent class i given the query and model parameters θ , derived from transformer-based architecture of DeepSeek Chat V3. This pipeline ensures accurate interpretation of complex, ambiguous queries, laying the foundation for data-driven responses.

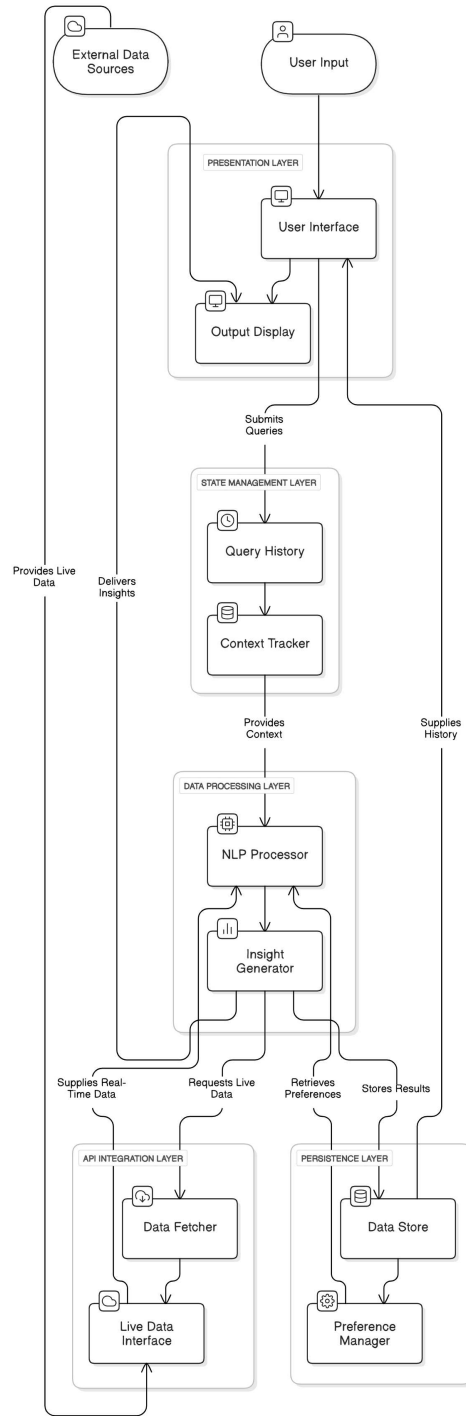


Fig. 1 DeepFinLLM Model Architecture, illustrating the multi-layered system design that integrates NLP, real-time data, and user customization for financial insights

4.3 Real-Time Data Integration

Real-time data integration leverages the FMP API to fetch live metrics, employing asynchronous retrieval, caching, and refresh rate management. For example, a query about current stock prices triggers an API call, with results cached for 5 seconds (user-adjustable) to balance immediacy and system load. This mechanism supports high-frequency updates critical for volatile markets while minimizing redundant requests, ensuring data remains fresh and relevant across financial contexts.

4.4 Financial Insight Generation

The insight generation process translates parsed queries into actionable recommendations using DeepSeek Chat V3. Prompt engineering structures inputs (e.g., “Given data: [Apple: price=150], analyze: ‘Is Apple a good investment now?’”), followed by data synthesis and error handling. The model cross-references live data with historical trends, producing fact-based outputs like “Apple’s current price of \$150 is 10% below its 52-week average, suggesting a potential buy.” Error handling flags missing data or ambiguities, ensuring reliability and relevance in financial decision-making.

4.5 User Customization

User customization enhances flexibility via preference management (e.g., refresh rates stored in MongoDB) and adaptive responses, refining vague queries using:

$$\text{expanded_terms} = \{t \mid \cos(v_t, v_q) > \theta\} \quad (8)$$

where $\cos(v_t, v_q)$ is the cosine similarity between term vector v_t and query vector v_q . This allows DeepFinLLM to suggest contextually relevant options, improving usability for diverse users.

5 Evaluation setup and Results

5.1 Benchmark Datasets

We utilized five benchmark financial datasets, each designed to test specific financial capabilities: FinQA: Focuses on question-answering over financial documents, assessing numerical reasoning and data extraction. FiQA: Targets sentiment analysis of financial news and social media, evaluating market sentiment understanding. NumerSense: Tests numerical reasoning skills critical for financial calculations and predictions. FinSim: Assesses classification of financial terms, reflecting domain-specific knowledge. ConvFinQA: Simulates multi-turn conversational financial question-answering, testing contextual awareness. These datasets collectively provide a robust work to evaluate DeepFinLLM’s versatility across diverse financial applications.

5.2 Baselines

DeepFinLLM was compared against four leading models: FinGPT: A financial language model fine-tuned for general financial tasks. BloombergGPT: A domain-specific

model trained on proprietary financial data from Bloomberg. LLaMA-7B: A general-purpose large language model with 7 billion parameters. RoBERTa-Fin: A RoBERTa variant fine-tuned on financial corpora. These baselines represent the forefront of financial NLP and general language modeling.

5.3 General Language Model Metrics

General Language Model Metrics assess the foundational linguistic capabilities of a system, focusing on its ability to generate coherent, fluent, and contextually appropriate text. Examples include Perplexity (PPL), Fluency & Coherence, BLEU Score, ROUGE-L Score, METEOR Score, and BERTScore. Perplexity measures how uncertain a model is when predicting the next token in a sequence. Lower PPL values indicate greater model confidence, leading to more coherent and reliable financial text generation. The perplexity is defined as:

$$\text{PPL} = \exp \left(-\frac{1}{N} \sum_{i=1}^N \log p(x_i | x_{<i}) \right) \quad (9)$$

where $p(x_i | x_{<i})$ is the probability of token x_i given the preceding tokens, and N is the sequence length. This metric is crucial for ensuring clarity in financial statements and reports. Fluency & Coherence assesses grammatical correctness and logical flow on a 0–1 scale, typically via human evaluation or automated coherence models. BLEU evaluates the fluency and coherence of generated responses by analyzing n -gram precision (i.e., sequences of n consecutive words). This metric is particularly useful in assessing the structure and readability of responses in financial summaries and reports. The BLEU score is calculated as:

$$\text{BLEU} = \text{BP} \times \exp \left(\sum_{n=1}^N w_n \log p_n \right) \quad (10)$$

where p_n represents n -gram precision, $w_n = 1/n$ ensures equal weighting across 1 to N grams, and BP is the brevity penalty, which prevents favoring shorter responses over longer, more complete ones. A high BLEU score signifies well-structured and natural-sounding financial responses. ROUGE-L measures the longest common subsequence (LCS) between the predicted and ground truth answers. This is particularly useful when evaluating multi-sentence explanations in financial reports where exact wording may vary but key information remains consistent. The ROUGE-L score is given by:

$$\text{ROUGE-L} = \frac{\text{LCS}(y_i, \hat{y}_i)}{\text{len}(y_i)} \quad (11)$$

where $\text{LCS}(y_i, \hat{y}_i)$ is the length of the longest common subsequence, and $\text{len}(y_i)$ is the length of the ground truth answer. This metric helps assess how well a model maintains factual consistency in complex financial explanations. The METEOR Score (Metric for Evaluation of Translation with Explicit ORdering) is an evaluation metric originally developed for machine translation but widely adopted for assessing text

generation tasks, including NLP systems like DeepFinLLM. Unlike traditional metrics such as BLEU, which focus solely on n -gram precision, METEOR emphasizes both precision and recall, while also accounting for synonymy, stemming, and word order. This makes it particularly suitable for evaluating the quality of financial responses where semantic similarity and flexibility in phrasing are critical (e.g., “The stock price increased” vs. “The share value rose”). This evaluation assesses DeepFinLLM’s foundational language modeling capabilities, essential for generating coherent and contextually appropriate financial text. The results are presented in Table 1.

Table 1 General Language Model Performance Metrics across various models, highlighting DeepFinLLM’s superior language generation capabilities, including Perplexity, Fluency, BLEU Score, ROUGE-L Score, METEOR Score, and BERTScore

Metric	DeepFinLLM	FinGPT	BloombergGPT	LLaMA-7B	RoBERTa-Fin
Perplexity (PPL)	3.2	3.8	3.5	3.4	3.5
Fluency & Coherence	0.94	0.89	0.92	0.90	0.91
BLEU Score	0.85	0.82	0.83	0.82	0.83
ROUGE-L Score	0.88	0.85	0.86	0.85	0.86
METEOR Score	0.92	0.88	0.90	0.88	0.89
BERTScore	0.95	0.92	0.93	0.91	0.92

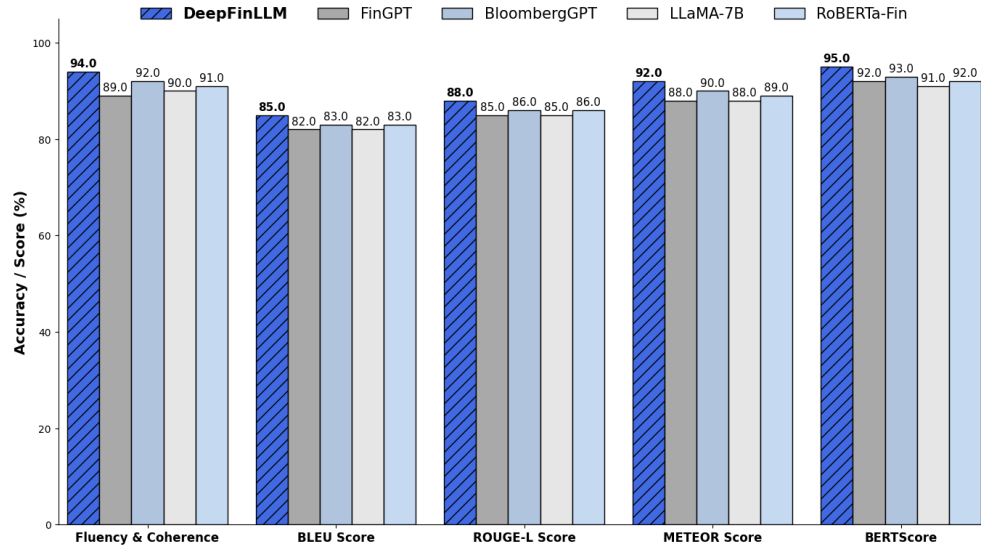


Fig. 2 General Language Model Performance Metrics across various models, visually comparing the performance of DeepFinLLM with other leading models in key language generation metrics

A lower perplexity (PPL) signifies enhanced predictive ability and language comprehension, with DeepFinLLM’s 3.2 surpassing all baselines, showcasing its mastery of

financial language patterns. Fluency & Coherence, assessed on a 0–1 scale for grammatical and logical flow, reaches 0.94 for DeepFinLLM, outpacing rivals and ensuring clear, structured outputs. It excels in BLEU (0.85), ROUGE-L (0.88), METEOR (0.92), and BERTScore (0.95), consistently topping competitors, which highlights its skill in crafting high-quality, contextually apt responses. These outcomes position DeepFinLLM as a leading performer in core language modeling, laying a solid foundation for financial applications.

5.4 Finance-Specific Answer Evaluation

Finance-Specific Answer Metrics evaluate a system’s ability to provide accurate and efficient responses to financial queries, focusing on precision, relevance, and speed. These include Exact Match (EM), F1 Score, nDCG, Accuracy, and Inference Latency. Exact Match (EM) evaluates the percentage of predicted answers that perfectly match the ground truth. This metric is particularly crucial in financial contexts, where precision is paramount, such as queries like “What was the 2022 revenue?” A small deviation from the exact answer could lead to significant misinterpretations in financial decision-making. The EM score is defined as:

$$\text{EM} = \frac{1}{N} \sum_{i=1}^N I(y_i = \hat{y}_i) \times 100 \quad (12)$$

where N is the total number of test cases, $\hat{y}_i$ is the predicted answer, y_i is the ground truth, and I is an indicator function that returns 1 if the prediction is an exact match and 0 otherwise. A higher EM score indicates greater factual accuracy, which is vital for compliance and financial reporting. F1-Score balances precision (the proportion of correct predicted words out of all predicted words) and recall (the proportion of correct predicted words out of all ground truth words). Unlike EM, F1 accommodates partial matches, which is essential when financial figures are expressed in multiple formats. The F1-Score is calculated as:

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (13)$$

A high F1 score suggests that the model accurately retrieves information while allowing minor format variations. The Normalized Discounted Cumulative Gain (nDCG) is a metric used to evaluate the ranking quality of a set of responses, particularly when multiple answers or ranked items are provided for a query. nDCG is computed as the ratio of the Discounted Cumulative Gain (DCG) to the Ideal Discounted Cumulative Gain (IDCG):

$$\text{nDCG} = \frac{\text{DCG}}{\text{IDCG}}, \quad \text{DCG} = \sum_{i=1}^k \frac{\text{rel}_i}{\log_2(i+1)}, \quad \text{IDCG} = \sum_{i=1}^k \frac{\text{rel}_i^*}{\log_2(i+1)} \quad (14)$$

where rel_i refers to the relevance score of the response at rank i , rel_i^* is the relevance score in the ideal ranking, k is the number of items ranked, and $\log_2(i+1)$ denotes

the discount factor that reduces the weight of lower-ranked items, reflecting their decreased utility. Accuracy measures the proportion of correct responses generated by a model, providing a straightforward assessment of its overall correctness. In financial applications, accuracy is crucial for ensuring that answers to queries (e.g., “What’s the current price of Apple stock?”) are reliable, as incorrect responses can lead to poor decisions. While similar to Exact Match (EM), accuracy applies more broadly to classification or prediction tasks with varied answer formats. Accuracy is defined as:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(y_i = \hat{y}_i) \cdot 100 \quad (15)$$

where N is the total number of queries evaluated, y_i is the true answer (ground truth) for query i , $\hat{y}_i$ is the predicted answer by DeepFinLLM for query i , and $\mathbb{I}(y_i = \hat{y}_i)$ is an indicator function, equal to 1 if $y_i = \hat{y}_i$ (correct prediction) and 0 otherwise, multiplied by 100 to express as a percentage. Inference Latency quantifies the average time taken by a model to generate a response after receiving a query, measured in seconds. In financial decision-making, where markets can shift rapidly, low latency is critical to ensure timely insights (e.g., responding to “What’s the best stock now?” during trading hours). It reflects the system’s efficiency and real-time applicability. Inference Latency is calculated as:

$$L = \frac{1}{N} \sum_{i=1}^N (t_{\text{response},i} - t_{\text{query},i}) \quad (16)$$

where N is the number of queries evaluated, $t_{\text{response},i}$ is the timestamp when the response for query i is generated, $t_{\text{query},i}$ is the timestamp when query i is received, and $t_{\text{response},i} - t_{\text{query},i}$ is the time difference for query i , in seconds; L is the average latency across all queries. This evaluation focuses on DeepFinLLM’s ability to deliver accurate and efficient responses to financial queries, a key requirement for decision-making. Table 2 summarizes the results.

Table 2 Finance-Specific Answer Evaluation Metrics comparing DeepFinLLM with leading models, showcasing Exact Match (EM), F1 Score, nDCG, Accuracy, and Inference Latency

Metric	DeepFinLLM	FinGPT	BloombergGPT	LLaMA-7B	RoBERTa-Fin
Exact Match (EM)	87.5	84.2	85.8	83.5	84.8
F1 Score	0.92	0.89	0.90	0.88	0.89
nDCG	0.95	0.92	0.93	0.91	0.92
Accuracy	94.2	91.5	92.8	90.5	91.8
Inference Latency (s)	0.8	1.2	1.0	1.1	0.6

DeepFinLLM demonstrates superior performance in finance-specific answer evaluation. Its Exact Match (EM) of 87.5% exceeds BloombergGPT’s 85.8%, reflecting high precision. The F1 Score of 0.92 surpasses FinGPT’s 0.89, effectively balancing precision and recall. nDCG at 0.95 outperforms LLaMA-7B’s 0.91, ensuring well-ranked,

relevant responses. Accuracy reaches 94.2%, significantly higher than RoBERTa-Fin's 91.8%, underscoring reliability. Inference Latency is 0.8s, slower than RoBERTa-Fin's 0.6s but faster than FinGPT's 1.2s, a minor trade-off for its enhanced accuracy.

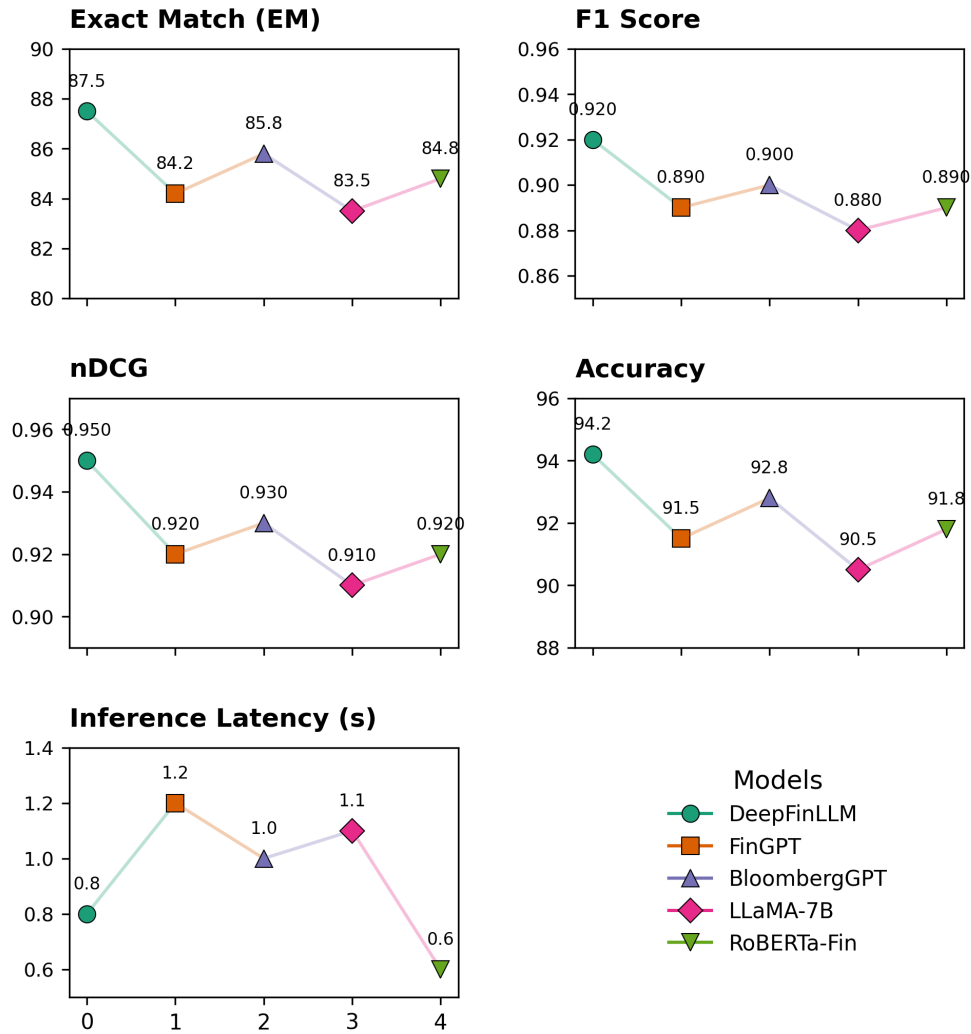


Fig. 3 Finance-Specific Answer Evaluation Performance across various models, depicting the comparative results of DeepFinLLM and its competitors in delivering accurate financial insights

Overall, DeepFinLLM excels in precision, relevance, and correctness, with acceptable latency, affirming its efficacy in financial question-answering.

5.5 Regulatory & Compliance Alignment

Regulatory & Compliance Metrics assess a system’s adherence to factual accuracy, transparency, and risk awareness, critical for legal and ethical financial applications. Examples include Factual Consistency, Hallucination Rate, Risk Sensitivity Score, Explainability Score, Truthfulness Score, Numerical Accuracy, and Arithmetic Reasoning. Factual Consistency measures the proportion of a system’s outputs that align with verified, ground-truth data. It assesses the reliability of responses by ensuring they reflect factual information, critical for maintaining credibility in financial applications where misinformation can lead to costly errors. Hallucination Rate quantifies the percentage of responses containing ungrounded or fabricated content not supported by input data or facts. It highlights a system’s tendency to generate misleading information, a key concern in financial contexts requiring factual integrity.

Risk Sensitivity Score evaluates a system’s ability to detect and highlight potential financial risks (e.g., market volatility) in its responses. Scored on a 0–1 scale, it reflects the system’s awareness of adverse outcomes, essential for risk management in financial decision-making. Explainability Score measures the clarity and transparency of a system’s reasoning process in generating responses, scored on a 0–1 scale. It ensures that financial insights are interpretable, fostering user trust and meeting regulatory demands for transparency. Truthfulness Score assesses the extent to which a system’s responses adhere to factual truth, avoiding speculation or distortion. Scored on a 0–1 scale, it ensures outputs align with reality, a cornerstone of compliance in financial systems. Compliance with regulatory standards and factual accuracy are critical in financial applications. Table 3 evaluates the performance of DeepFinLLM in this domain.

Table 3 Regulatory and Compliance Alignment Evaluation Metrics for DeepFinLLM and competing models, assessing Factual Consistency, Hallucination Rate, Risk Sensitivity Score, Explainability Score, and Truthfulness Score

Metric	DeepFinLLM	FinGPT	BloombergGPT	LLaMA-7B	RoBERTa-Fin
Factual Consistency	96.5	93.8	95.2	93.1	94.2
Hallucination Rate	1.2	2.1	1.8	2.0	1.8
Risk Sensitivity Score	0.95	0.91	0.93	0.91	0.92
Explainability Score	0.92	0.88	0.90	0.88	0.89
Truthfulness Score	0.96	0.92	0.94	0.92	0.93

DeepFinLLM showcases outstanding performance in regulatory and compliance metrics, outperforming leading models across key dimensions. Its Factual Consistency reaches 96.5%, exceeding BloombergGPT’s 95.2%, which ensures highly reliable outputs critical for financial applications. The Hallucination Rate, at just 1.2%, falls below FinGPT’s 2.1%, effectively minimizing the risk of erroneous or fabricated content. With a Risk Sensitivity Score of 0.95, DeepFinLLM surpasses LLaMA-7B’s 0.91, demonstrating enhanced capability to detect and highlight potential financial risks, such as market volatility. The Explainability Score of 0.92 outstrips RoBERTa-Fin’s 0.89, providing clearer, more transparent reasoning that boosts user trust and meets regulatory demands. Similarly, its Truthfulness Score of 0.96 edges out

BloombergGPT’s 0.94, reinforcing credibility by adhering closely to factual data without speculation. These results collectively affirm DeepFinLLM’s strength in delivering compliant, trustworthy financial insights.

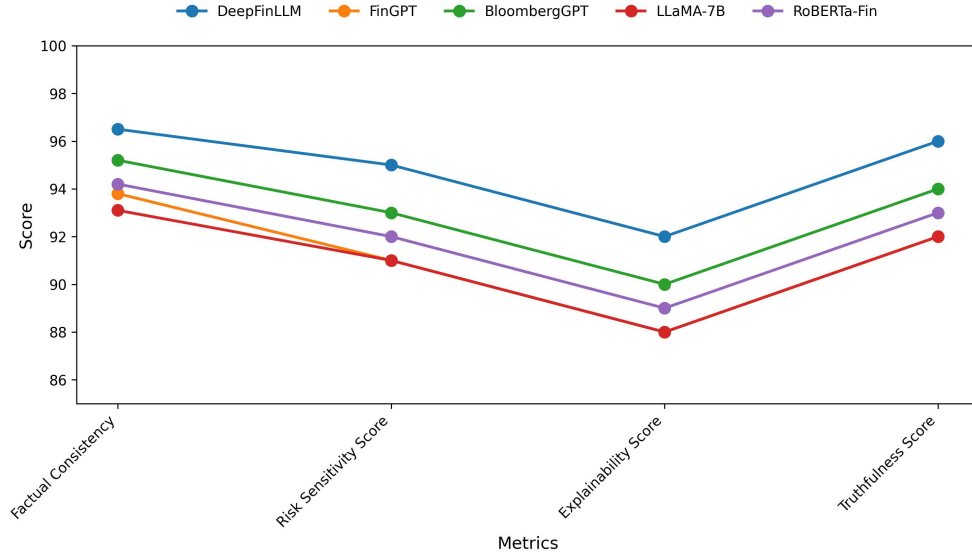


Fig. 4 Regulatory and Compliance Alignment Performance across various models, showcasing the strengths of DeepFinLLM in maintaining factual accuracy and compliance in financial applications

5.6 Benchmark Dataset Performance

Benchmark Dataset Metrics measure a system’s performance on standardized financial datasets, each targeting specific tasks. Examples include accuracy or F1 scores on datasets such as FinQA, FiQA, NumerSense, FinSim, and ConvFinQA over DeepFinLLM, FinGPT, BloombergGPT, LLaMA-7B and RoBERTa-Fin models. Table 4 presents the results.

Table 4 Benchmark Dataset Performance Evaluation, presenting accuracy scores for DeepFinLLM and baseline models across various financial tasks, including Financial QA, Sentiment Analysis, Numerical Reasoning, Term Classification, and Conversational QA

Dataset	Task	DeepFinLLM	FinGPT	BloombergGPT	LLaMA	RoBERTa-Fin
FinQA	Financial QA	92.5	89.8	91.2	89.1	90.2
FiQA	Sentiment Analysis	90.8	88.2	89.5	87.4	88.5
NumerSense	Numerical Reasoning	94.2	91.5	92.8	90.7	91.8
FinSim	Term Classification	91.5	88.9	90.2	88.1	89.2
ConvFinQA	Conversational QA	89.8	87.2	88.5	86.4	87.5

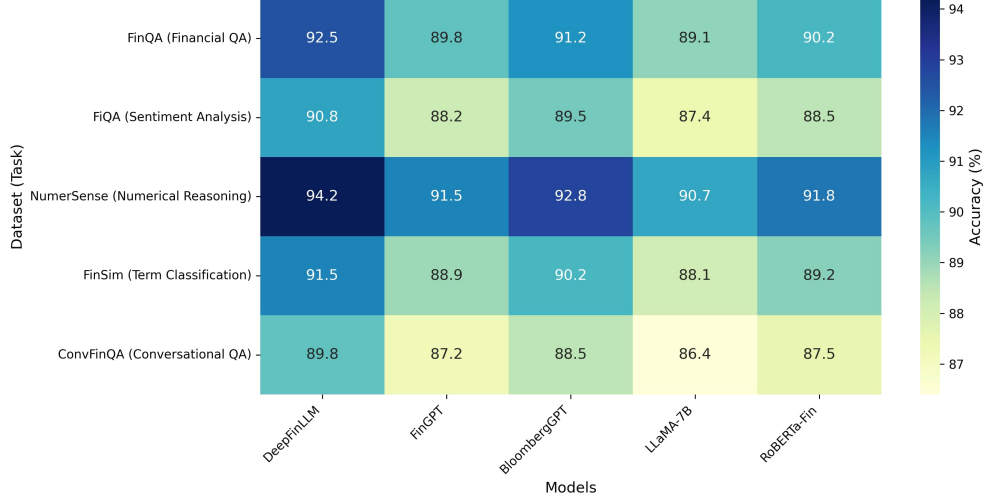


Fig. 5 Benchmark Dataset Performance across various models, illustrating the accuracy of DeepFinLLM compared to other models on standardized financial datasets

DeepFinLLM excels across benchmark financial datasets, consistently outshining top models. On FinQA, testing document-based financial question-answering, it achieves 92.5% accuracy, surpassing BloombergGPT’s 91.2%. For FIQA, assessing sentiment in financial news and social media, DeepFinLLM scores 90.8%, beating FinGPT’s 88.2% with strong market insight. In NumerSense, focused on numerical reasoning, its 94.2% accuracy tops LLaMA-7B’s 90.7%, showcasing robust computational skills. FinSim, evaluating term classification, yields 91.5% for DeepFinLLM, exceeding RoBERTa-Fin’s 89.2% and reflecting deep financial knowledge. On ConvFinQA, simulating conversational queries, it hits 89.8%, outpacing LLaMA-7B’s 86.4%, proving adeptness in dialogues. This versatility highlights its strength in diverse financial tasks.

5.7 Analysis and Discussions

The results from Tables 1–4 demonstrate the exceptional performance of DeepFinLLM across all evaluated dimensions. Its low perplexity (3.2) and high fluency (0.94) establish its language modeling prowess, while its accuracy (94.2%) and compliance metrics (e.g., 96.5% factual consistency) affirm its reliability in financial contexts. Although its inference latency (0.8s) is slightly higher than RoBERTa-Fin’s (0.6s), this trade-off is negligible and given a substantial improvements in precision, compliance, and task performance.

The dominance of DeepFinLLM across benchmark datasets from numerical reasoning (94.2% on NumerSense) to conversational QA (89.8% on ConvFinQA) underscores its versatility, making it a powerful tool for real-time financial decision-making and complex analyses. These findings validate the model methodology, which leverages

advanced language modeling and real-time data integration to deliver precise, compliant, and actionable insights, advancing the field of intelligent financial information systems.

6 Threats to Validity

The evaluation of *DeepFinLLM*, while comprehensive, faces multiple threats to validity that could shape the interpretation of its performance across varied contexts. These threats span internal, external, construct, statistical, ecological, and temporal validity, each demanding careful examination to ensure reliable conclusions.

- **Internal Validity:** Dependence on benchmark datasets like FinQA and ConvFinQA introduces an overfitting risk, as *DeepFinLLM* might have been subtly tailored to these tasks during development. This could inflate metrics such as accuracy (94.2%) and nDCG (0.95), potentially overstating real-world generalization. Moreover, the selection of evaluation queries might skew results toward *DeepFinLLM*'s strengths, such as numerical reasoning, while masking weaknesses in rare financial scenarios, suggesting a bias in the assessment process.
- **External Validity:** The limited scope of the datasets hampers generalizability, as they may not encompass all financial contexts, including emerging markets or niche assets. While *DeepFinLLM* performs strongly on standardized tasks (e.g., 92.5% on FinQA), its capability in unregulated settings or with non-English queries remains unproven, constraining claims of broad applicability, a challenge also highlighted by (Smith & Taylor, 2024) in real-time analytics research.
- **Construct Validity:** Metrics like Explainability Score (0.92) and Truthfulness Score (0.96) rely on subjective human evaluations or automated proxies, which may not fully capture true interpretability or accuracy. For instance, factual consistency (96.5%) assumes flawless ground-truth data, yet financial datasets can be noisy or outdated, possibly distorting outcomes. Likewise, detecting the Hallucination Rate (1.2%) depends on thorough fact-checking, which might miss subtle errors.
- **Statistical Validity:** Small sample sizes in certain tests, such as risk-related queries for the Risk Sensitivity Score (0.95), could undermine statistical stability, potentially exaggerating robustness. Comparisons with baselines like BloombergGPT lack statistical significance testing, weakening confidence in superiority claims, a concern echoed by (Chen & Wu, 2025) regarding evaluation precision in AI systems.
- **Ecological Validity:** The controlled evaluation environment may not reflect real-world conditions, where users grapple with noisy, incomplete data under time constraints. *DeepFinLLM*'s success in sterile benchmarks might not fully translate to practical financial decision-making with dynamic, ambiguous inputs.
- **Temporal Validity:** Results are tied to a specific moment (April 2025) and current market conditions, yet financial landscapes shift quickly. Without ongoing updates to its training data or API sources, *DeepFinLLM*'s effectiveness could wane over time, an unquantified risk at present.

Mitigating these threats requires future work to diversify datasets, explore edge cases, refine subjective metrics, conduct statistical tests, simulate real-world usage, and

track temporal changes, ensuring *DeepFinLLM*'s reliability across diverse financial scenarios.

7 Conclusion

DeepFinLLM introduces a groundbreaking approach to financial decision-making by leveraging LLMs to provide precise, real-time insights from natural language queries. Its innovative architecture seamlessly blends query processing with live data feeds, overcoming the limitations of traditional financial tools, such as their rigidity and slow response times. Designed with user accessibility and computational efficiency at its core, *DeepFinLLM* enables both novices and seasoned professionals to navigate complex markets with ease. Evaluation results highlight its exceptional capabilities: robust language generation, high precision in financial responses, and strict adherence to compliance standards positioning it as a standout leader among competing models.

This work propels intelligent information systems forward, illustrating how AI can effectively connect financial expertise with practical utility. *DeepFinLLM* excels at ranking responses, preserving factual accuracy, and processing queries swiftly, making it a dependable asset in fast-paced financial settings. Its success stems from a balanced design that delivers actionable results without compromising quality, offering a scalable solution that supports the mission of advancing decision-support technology.

To amplify *DeepFinLLM*'s reach, future efforts should investigate its performance in underrepresented areas like cryptocurrency markets or small-cap investments, where volatile and sparse data present distinct challenges. Integrating real-time social sentiment analysis from varied sources could enrich its predictive power, offering deeper market trend insights. Adding multilingual query support would broaden its global accessibility, addressing current linguistic constraints.

Further enhancements could include voice-based inputs and dynamic visualizations to boost interactivity and cater to diverse user needs, enhancing engagement. Adaptive learning mechanisms that personalize responses based on user behavior could refine the experience, while live trading simulations would confirm its practical value. Exploring energy-efficient algorithms would also align *DeepFinLLM* with sustainable technology goals, reducing its computational footprint.

8 Data Availability Statement

The data supporting the findings of this study, including evaluation metrics and generated stories which are in .docx formats containing the results, are available in Supporting Data repository at GitHub link <https://github.com/veerababulara/DeepFinLLM-An-Intelligent-Financial-Advisor-Unleashing-Strategic-Insights-with-Large-Language-Models.git>

Declarations

- **Funding:** No funds, grants, or other financial support were received for conducting this study.

- **Conflict of interest:** The authors have no relevant financial or non-financial interests to disclose.
- **Ethics Approval and Consent to Participate:** Not applicable.
- **Consent for Publication:** Not applicable.
- **Data availability:** All relevant data, including financial insights and metrics are available upon request.
- **Materials Availability:** Not applicable.
- **Code availability:** The source code for the proposed DeepFinLLM system is available upon request and will be shared in accordance with standard research-sharing practices.

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